Industrial Internship Report on

” Voice-Controlled Robotic Vehicle for Autonomous Navigation”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was (“Voice-Controlled Robotic Vehicle for Autonomous Navigation.” The main objective of the project was to design and develop a multifunctional robotic vehicle capable of operating in different modes such as line following, obstacle avoidance, voice control, and manual mobile app control. The system was developed using Arduino UNO, IR sensors, ultrasonic sensor, Bluetooth module, motor driver, servo motor, and a mobile application interface. The robot can autonomously navigate by following a predefined path, avoid obstacles using ultrasonic sensing, and also respond to user voice commands and mobile app controls through Bluetooth communication. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

The six-week Industrial Internship conducted by Upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT) was a highly valuable and informative learning experience. During this internship, we worked on understanding industrial requirements, planning project execution, designing hardware and software modules, testing system performance, and preparing the final technical report. The internship provided practical exposure to real-world engineering applications and helped us improve our technical and problem-solving skills.In today's competitive world, internships play an important role in career development because they bridge the gap between theoretical knowledge and practical implementation. Through this internship, we understood how industrial projects are developed, managed, and completed within a limited timeline. It also improved our communication skills, teamwork, project management abilities, and technical documentation skills, which are essential for future professional growth.

Our project, "Voice-Controlled Robotic Vehicle for Autonomous Navigation," focused on developing a multifunctional robotic system capable of operating in multiple modes such as line following, obstacle avoidance, voice control, and manual mobile app control. The robot was developed using Arduino UNO, IR sensors, ultrasonic sensor, HC-05 Bluetooth module, servo motor, and L298N motor driver. The project aimed to create a low-cost and efficient robotic platform capable of autonomous as well as user-controlled navigation.

The opportunity provided by Upskill Campus (USC), The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) was highly beneficial. The internship program was systematically planned with regular guidance sessions, project reviews, technical learning activities, and report preparation support. Weekly milestones and continuous mentoring helped us complete the project successfully within the six-week duration.

Throughout this internship, we gained knowledge in embedded systems, robotics, Arduino programming, sensor interfacing, Bluetooth communication, automation, and mobile app integration. We also learned how to troubleshoot hardware and software issues, improve system performance, and work effectively as a team. Overall, the internship enhanced our confidence, technical knowledge, and practical implementation skills.

We sincerely thank Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) for providing us with this valuable opportunity. We are grateful to our mentors, project guides, faculty members, and coordinators for their continuous support and encouragement throughout the internship. We would also like to thank our teammates, friends, and family members who helped us directly or indirectly during the successful completion of the project.

Finally, we would like to encourage our juniors and peers to actively participate in internships and practical projects. Such opportunities provide real industry exposure, improve technical knowledge, and help build confidence for future careers. Continuous learning, innovation, and practical implementation are the keys to success in the field of engineering and technology.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

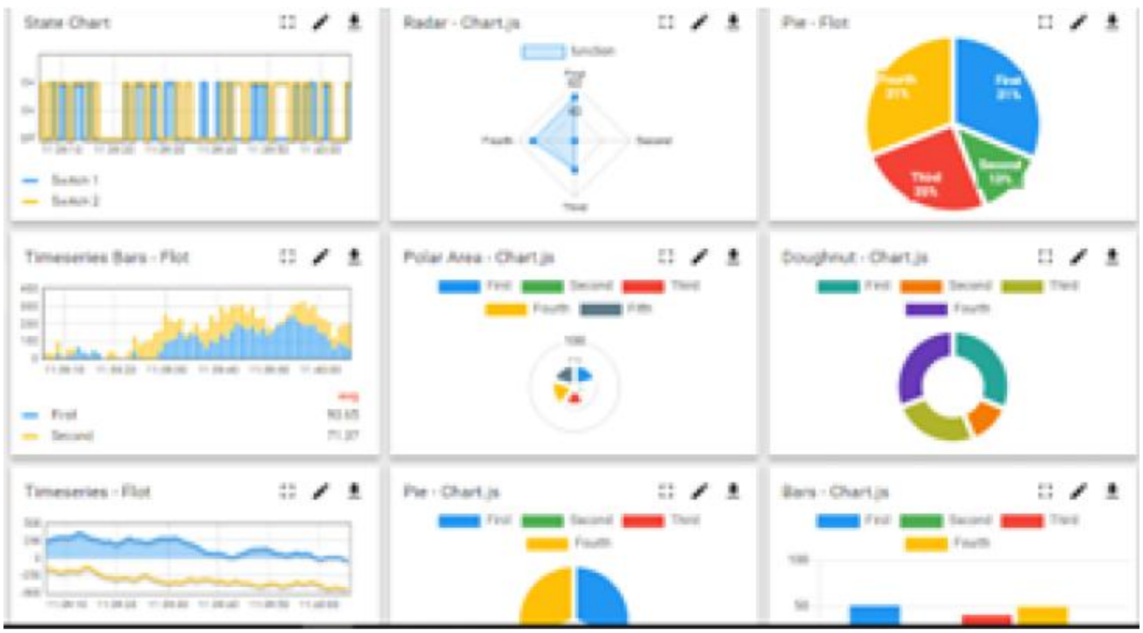
i. UCT IoT Platform ( )

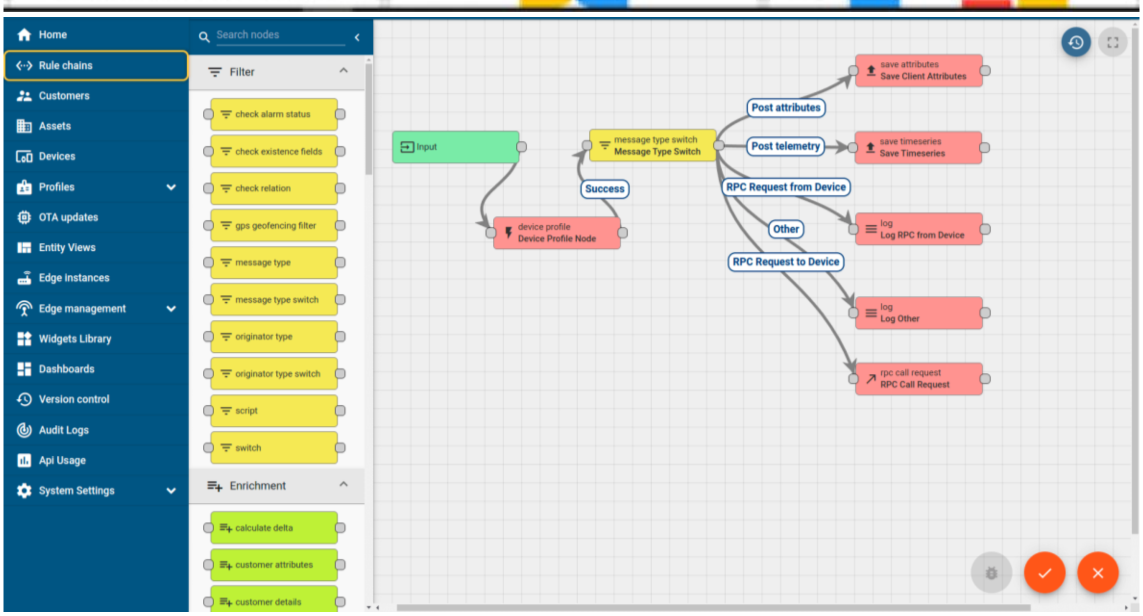
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

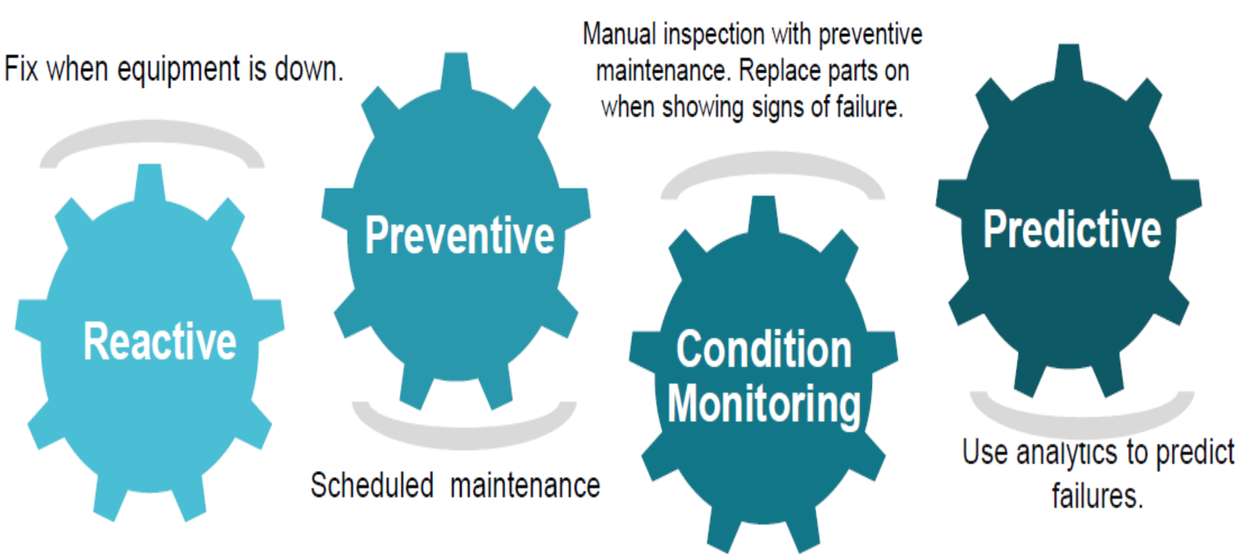
Its unique SaaS model helps users to save time, cost and money.

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

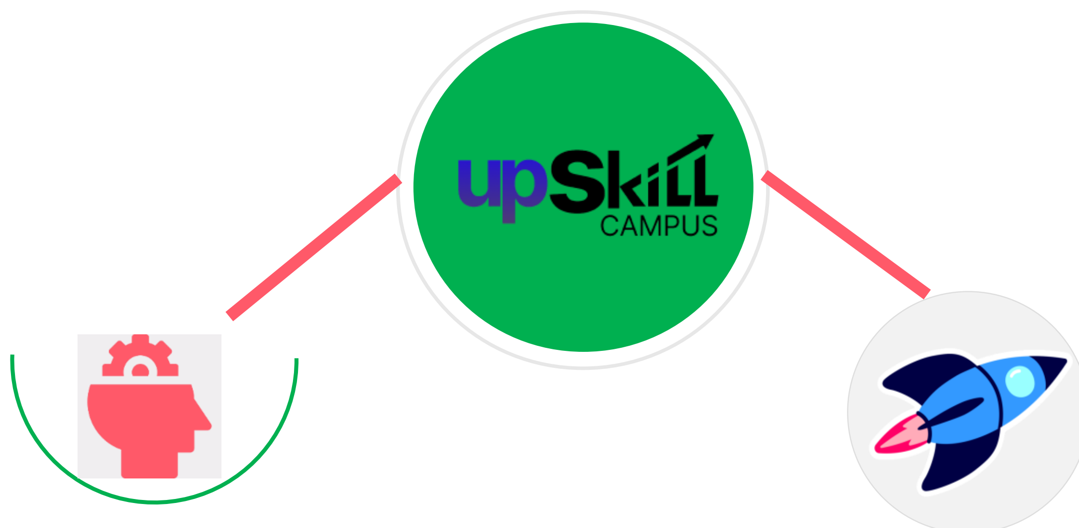
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

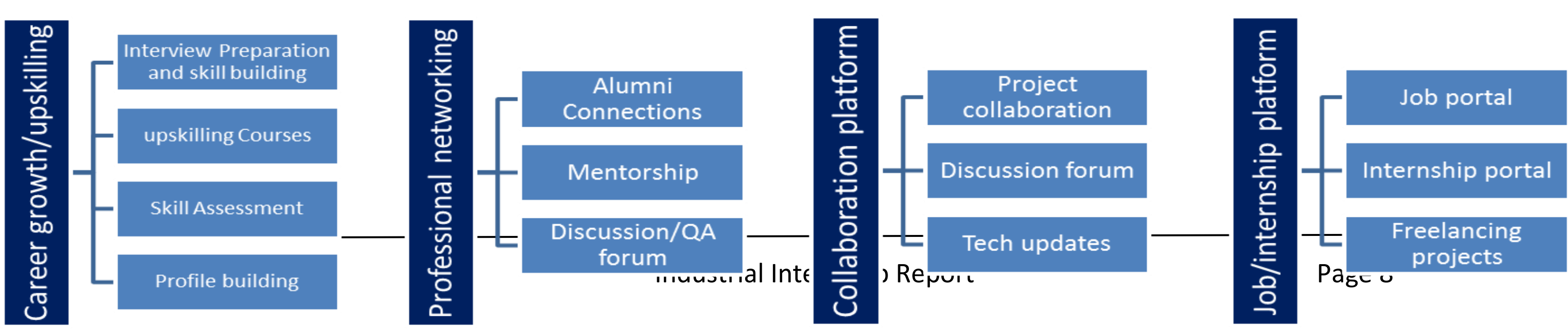
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Arduino UNO Documentation – https://www.arduino.cc/

[2] HC-05 Bluetooth Module Datasheet

[3] HC-SR04 Ultrasonic Sensor Datasheet

[4] MIT App Inventor Official Documentation – https://appinventor.mit.edu/

[5] L298N Motor Driver Module Documentation

2.6 Glossary

Terms	Acronym
Arduino Integrated Development Environment	Arduino IDE
Infrared Sensor	IR Sensor
Ultrasonic Sensor	HC-SR04
Bluetooth Module	HC-05
Motor Driver Module	L298N

3 Problem Statement

In the assigned problem statement, the objective was to design and develop a low-cost multifunctional robotic vehicle capable of autonomous and manual navigation. Traditional robotic systems are generally designed to perform only a single operation such as line following, obstacle avoidance, or remote control. This increases system cost, hardware complexity, and reduces operational flexibility.

The problem addressed in this project was to integrate multiple functionalities into a single robotic platform capable of performing line following, obstacle avoidance, voice-controlled operation, and manual mobile app control. The robot needed to operate efficiently using sensor-based decision making and wireless communication while maintaining reliable movement and control.

The project aimed to develop a practical robotic system using Arduino UNO, IR sensors, ultrasonic sensor, Bluetooth module, motor driver, and mobile application integration. The system was required to provide accurate navigation, obstacle detection, smooth motor control, and user interaction through both voice commands and manual controls.

The main goal of this project was to create an intelligent and flexible robotic vehicle suitable for educational, automation, and real-world robotics applications while improving understanding of embedded systems and autonomous navigation technologies.

4 Existing and Proposed solution

Existing robotic systems developed by others generally focus on performing a single specific task such as line following, obstacle avoidance, or Bluetooth-based manual control. Many existing robots use separate hardware setups for each operation, which increases system complexity, cost, and maintenance requirements. Some systems provide autonomous navigation but lack user interaction features such as voice control or mobile app control. Other robots depend heavily on expensive sensors and advanced processors, making them less suitable for educational and low-cost applications.

To overcome these limitations, the proposed solution is the development of a “Voice-Controlled Robotic Vehicle for Autonomous Navigation.” The proposed robot integrates multiple functionalities into a single compact platform. The system combines:

• Line following using IR sensors
• Obstacle avoidance using ultrasonic sensor and servo motor
• Voice command control using Bluetooth communication
• Manual mobile app control through Android application

The robot is developed using Arduino UNO, L298N motor driver, HC-05 Bluetooth module, ultrasonic sensor, IR sensors, and DC motors. The system can autonomously navigate and also allow user interaction through voice and mobile controls.

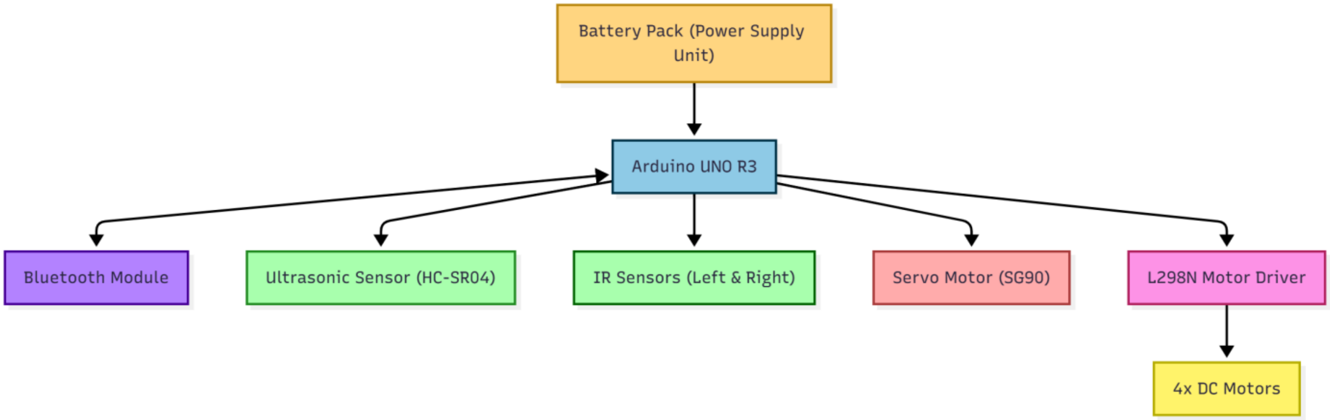
The value additions planned in this project include:

• Multi-functional operation within a single robotic platform
• Low-cost and easy-to-develop system design
• Improved user interaction through voice commands
• Flexible switching between autonomous and manual modes
• Educational and industrial learning applications
• Expandability for future technologies such as IoT, AI, GPS, and camera-based navigation
• Improved practical understanding of robotics and embedded systems

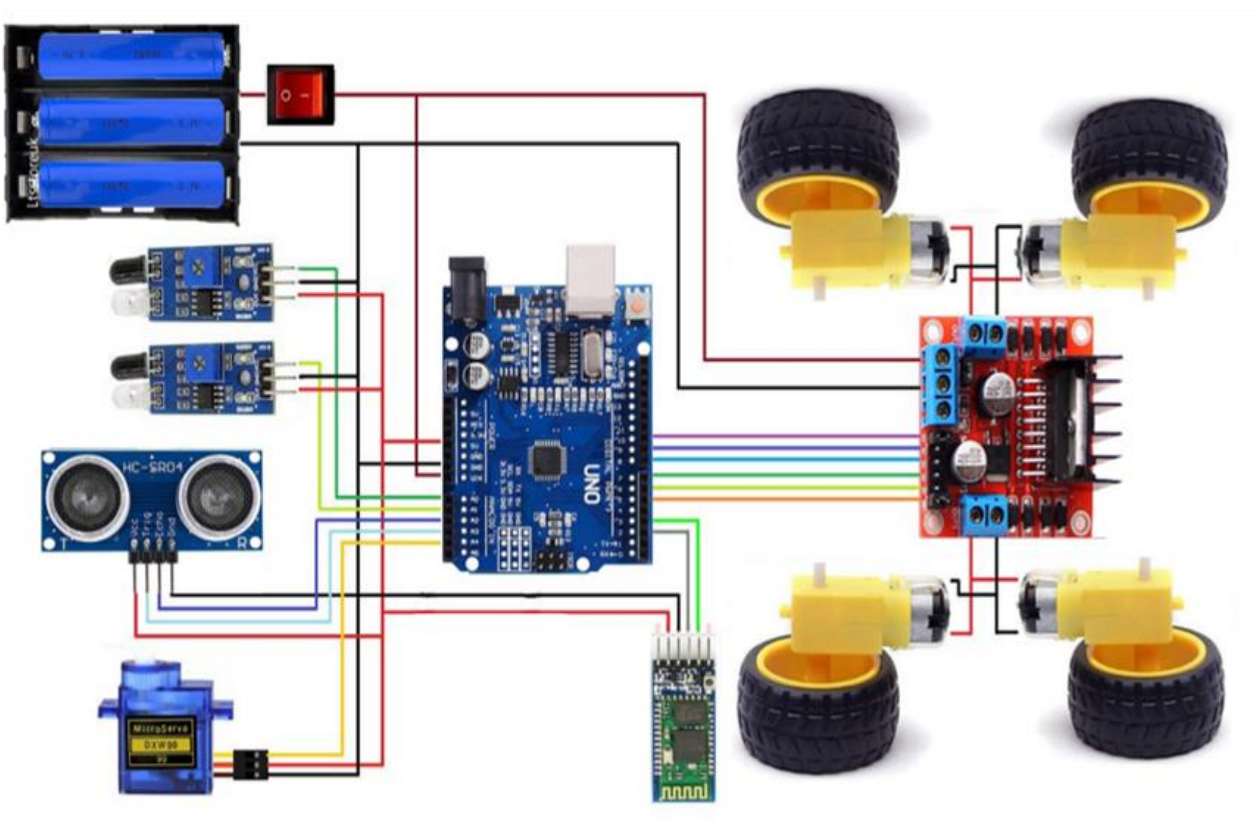
4.1 Code submission (upskillcampus/VoiceControlledRoboticVehicle.ino at main · Tirtha2005/upskillcampus)

4.2 Report submission (upskillcampus/VoiceControlledRoboticVehicle_Tirtha Rudra Shettigar_USC_UCT.pdf at main · Tirtha2005/upskillcampus)

5 Proposed Design/ Model

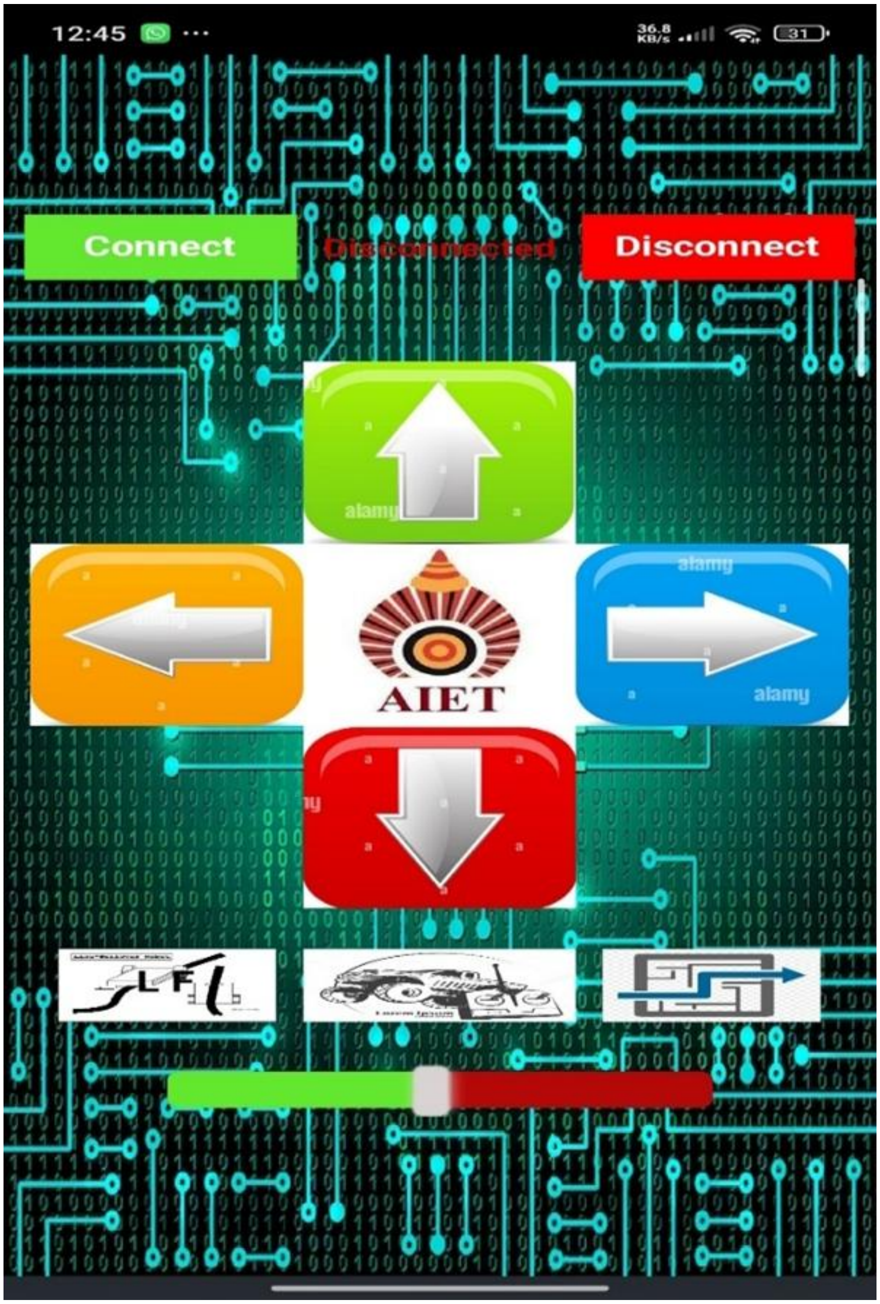


5.1 High Level Diagram (if applicable)



5.2 Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)



6 Performance Test

The performance testing of the "Voice-Controlled Robotic Vehicle for Autonomous Navigation" was carried out to evaluate the efficiency, accuracy, and reliability of the system under different operating conditions. The project was tested in line following, obstacle avoidance, voice control, and manual mobile control modes.

Some important constraints identified during testing were sensor accuracy, Bluetooth communication range, battery backup, motor response time, and environmental noise during voice control. These constraints were minimized by proper sensor calibration, stable power supply, optimized Arduino programming, and efficient motor control using PWM signals.

The IR sensors provided stable line tracking on smooth surfaces, while the ultrasonic sensor accurately detected obstacles within the specified range. Bluetooth communication remained stable within approximately 8–10 meters. Voice command performance depended on environmental noise conditions and command clarity. The battery backup supported continuous operation for around 30–40 minutes.

6.1 Test Plan/ Test Cases

• Test line-following accuracy on straight and curved paths
• Test obstacle detection and automatic turning response
• Test Bluetooth connectivity range and response speed
• Test voice command recognition and robot movement
• Test manual mobile app control functionality
• Test battery performance during continuous operation

6.2 Test Procedure

• The robot was placed on a black line track to test line-following capability.
• Obstacles were positioned at different distances to verify ultrasonic sensing and avoidance.
• Bluetooth communication was tested using an Android mobile application.
• Voice commands such as "forward," "left," "right," and "stop" were given through the app.
• Continuous operation testing was performed to evaluate system stability and battery performance.

6.3 Performance Outcome

• Line-following accuracy achieved was approximately 90% on standard tracks.
• Obstacle avoidance worked successfully for objects detected within 20 cm range.
• Bluetooth communication remained stable up to 8–10 meters.
• Voice command accuracy was approximately 85–90% under normal conditions.
• The robot operated continuously for around 30–40 minutes using battery power.
• Overall system performance was stable and reliable across all operating modes.

7 My learnings

This internship provided me with valuable practical knowledge and real-time industry exposure in the field of robotics, embedded systems, and automation. During the development of the "Voice-Controlled Robotic Vehicle for Autonomous Navigation," I learned how to integrate hardware and software components to build a fully functional robotic system.

I gained hands-on experience in Arduino programming, sensor interfacing, Bluetooth communication, motor driver control, and mobile app integration. I also learned how IR sensors are used for line following, how ultrasonic sensors help in obstacle detection, and how voice commands can be implemented using Bluetooth communication.

Apart from technical skills, this internship improved my problem-solving ability, teamwork, communication skills, project planning, and technical documentation skills. Working within a fixed timeline helped me understand the importance of time management and systematic project execution.

This internship will greatly help in my career growth by improving my practical understanding of embedded systems, IoT, and robotics technologies. The experience gained through this internship has increased my confidence in developing real-world engineering projects and prepared me for future opportunities in automation, robotics, and software-hardware integration fields.

8 Future work scope

The current project successfully demonstrates line following, obstacle avoidance, voice control, and manual mobile app control. However, several advanced features can be added in the future to improve the robot's intelligence, efficiency, and real-world applications.

Future improvements may include:
• Integration of IoT modules such as ESP8266 or ESP32 for internet-based control and monitoring
• Addition of camera modules for image processing and object recognition
• Implementation of GPS-based navigation for outdoor autonomous movement
• Use of AI and Machine Learning algorithms for intelligent decision making
• Integration of LiDAR or advanced sensors for more accurate obstacle detection
• Development of real-time video streaming through the mobile application
• Solar-powered battery charging system for longer operation
• Multi-robot communication and swarm robotics applications
• Improved mobile application with gesture control and live sensor monitoring

These enhancements can make the robotic system more advanced, intelligent, and suitable for industrial, surveillance, automation, and smart robotics applications.